

COMPUTER VISION MIDTERM and FINAL PROJECT

Master Guide Index - START HERE!

Welcome to your Computer Vision project! This master guide will help you navigate all the project documentation.

Quick Start - What Should I Read First?

If you're just starting:

1. **Read this page** (you are here!)
2. **Getting Started Guide** - Step-by-step for beginners
3. **Project Ideas & Tiers** - Choose your project
4. **Weekly Checklist** - Track your progress

If you're planning your proposal:

1. **Midterm Guide** - Proposal requirements
2. **Project Ideas & Tiers** - Pick your tier
3. **Resources & Setup** - Get your environment ready

If you're implementing:

1. **Getting Started Guide** - Week-by-week instructions
2. **Resources & Setup** - Code templates and tutorials
3. **AI Usage Policy** - How to use AI tools properly

If you're preparing final submission:

1. **Final Project Guide** - Final requirements
 2. **Weekly Checklist** - Final week checklist
 3. **FAQ** - Common concerns
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Document Overview

1. MIDTERM - Project Proposal Guide

File: `1_Midterm_Proposal.md`

What's in it:

- Midterm objectives and deliverables
- Presentation structure (8-10 slides)
- GitHub repository setup
- README template
- Grading rubric (100 points)
- Submission checklist

When to use it:

- Week 9-10 (Oct 16 - Oct 30)
- When preparing your proposal
- Before midterm presentation

Key sections:

- Presentation slide requirements
 - GitHub structure
 - README template
 - Grading criteria
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2. FINAL - Complete Project Guide

File: `2_Final_Project.md`

What's in it:

- Final project requirements
- Complete repository structure
- README template (final version)
- Demo video requirements (3-5 min)
- Presentation requirements (10 min)
- Grading rubric (100 points + bonus)

When to use it:

- Week 11-15 (Nov 6 - Dec 4)
- When implementing your project
- Before final presentation

Key sections:

- Repository structure
 - Demo video format
 - Live presentation requirements
 - Final grading criteria
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3. Project Ideas & Tiers

File: `3_Project_Ideas.md`

What's in it:

- How to choose your tier
- 8 Tier 1 project ideas (★ ★ ☆ ☆ ☆)
- 8 Tier 2 project ideas (★ ★ ★ ☆ ☆)
- 8 Tier 3 project ideas (★ ★ ★ ★ ★)
- Project difficulty ratings
- Resource requirements for each tier

When to use it:

- Week 9 (Oct 16-23)
- When choosing your project
- When deciding on project scope

Key sections:

- Tier comparison
 - Specific project examples
 - Resource needs per tier
 - Decision framework
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4. Getting Started Guide for Beginners

File: `4_Getting_Started.md`

What's in it:

- Week-by-week roadmap
- Day-by-day tasks
- Code examples with explanations
- Troubleshooting common issues
- Step-by-step setup instructions

When to use it:

- Week 9-15 (entire project)
- When you're stuck
- As your main reference
- For implementation help

Key sections:

- Weekly task breakdown
 - Code templates
 - Environment setup
 - Debugging tips
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5. AI Assistant Usage Policy

File: `5_AI_Usage_Policy.md`

What's in it:

- What AI usage is allowed (and encouraged!)
- What AI usage is not allowed
- AI usage log requirements
- Log template with examples
- How to document AI usage

When to use it:

- Throughout entire project
- Before using ChatGPT/Claude/Copilot
- When documenting AI assistance
- Before submission

Key sections:

- Allowed vs not allowed usage
 - AI usage log template
 - Verification criteria
 - Best practices
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6. Resources & Technical Setup

File: `6_Resources_Setup.md`

What's in it:

- Free GPU resources (Colab, Kaggle, Heidi)
- Free APIs and services
- Pre-trained model sources
- Dataset sources
- Code templates
- Software installation guides
- Troubleshooting technical issues

When to use it:

- Week 9 (setup phase)
- When choosing resources
- When looking for datasets
- When you have technical issues

Key sections:

- GPU options comparison
 - Dataset repositories
 - Code templates
 - Installation guides
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7. Frequently Asked Questions (FAQ)

File: `7_FAQ.md`

What's in it:

- Common questions organized by topic:
 - About the project
 - About resources
 - About teams
 - About grading
 - About AI usage
 - About data
 - Technical questions
 - Timeline questions

When to use it:

- Anytime you have a question!
- Before asking instructor
- When stuck or confused
- For quick answers

Key sections:

- Organized by topic
 - Clear answers
 - No judgment questions
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8. Weekly Checklist & Timeline

File: 8_Weekly_Checklist.md

What's in it:

- Project timeline overview
- Week-by-week checklists
- Daily task breakdowns
- Progress tracking templates
- Warning signs if you're behind
- Pre-submission final checklist

When to use it:

- Every week throughout project
- Start of each week (plan)
- End of each week (review)
- When tracking progress

Key sections:

- Weekly checklists
 - Daily tasks
 - Progress tracking
 - Recovery plans
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9. Quick Reference Card

File: 9_Quick_Reference_Card.md

What's in it:

- Critical dates
- Grading breakdown
- Project tiers summary
- What to submit
- Free resources
- Weekly goals
- AI usage rules
- Emergency contacts

When to use it:

- PRINT THIS!
- Daily reference
- Quick lookup

- Desktop reminder

Key sections:

- One-page summary
 - Essential information
 - Quick troubleshooting
 - Important links
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10. GitHub Templates

File: `10_GitHub_Templates.md`

What's in it:

- Complete repository structure
- README.md templates (midterm & final)
- requirements.txt
- .gitignore
- data/README.md
- models/README.md
- AI_usage_log.md (full example)

When to use it:

- Week 9-10 (setup)
- When creating GitHub repo
- When documenting project
- Copy-paste ready files

Key sections:

- File templates
 - Example content
 - Structure guidelines
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How to Navigate the Docs

By Week:

Week 9 (Oct 16-23): → Start with: Getting Started Guide (Week 9) → Then read: Project Ideas & Tiers → Also review: Resources & Setup

Week 10 (Oct 30): → Focus on: Midterm Proposal Guide → Use: Weekly Checklist (Week 10)
→ Remember: AI Usage Policy

Weeks 11-14: → Follow: Getting Started Guide (respective weeks) → Reference: Resources & Setup (as needed) → Check: Weekly Checklist (each week) → Document: AI Usage Policy (ongoing)

Week 15 (Dec 4): → Focus on: Final Project Guide → Follow: Weekly Checklist (Week 15) → Double-check: FAQ (if concerns)

By Task:

Choosing a project: → Project Ideas & Tiers

Setting up environment: → Resources & Setup → Getting Started Guide

Writing proposal: → Midterm Proposal Guide

Implementing code: → Getting Started Guide → Resources & Setup

Using AI tools: → AI Usage Policy

Creating demo: → Getting Started Guide (Week 13) → Final Project Guide

Preparing presentation: → Final Project Guide → Weekly Checklist (Week 14-15)

Stuck or confused: → FAQ → Getting Started Guide (Troubleshooting)

By Role:

Complete Beginner:

1. Getting Started Guide (read fully)
2. Project Ideas & Tiers (choose Tier 1)
3. Weekly Checklist (follow religiously)
4. FAQ (keep handy)

Experienced Student:

1. Project Ideas & Tiers (consider Tier 2/3)
2. Midterm/Final Guides (know requirements)
3. Resources & Setup (pick advanced tools)
4. AI Usage Policy (document well)

Team Leader:

1. Weekly Checklist (coordinate team)
2. All guides (understand full scope)
3. FAQ (help teammates)

Struggling Student:

1. FAQ (find quick answers)
2. Getting Started Guide (step by step)
3. Weekly Checklist (get back on track)
4. Contact instructor!



Important Dates & Milestones

Date	Event	Documents to Reference
Oct 16	Project Kickoff	Getting Started (Week 9)
Oct 23	NO CLASS	Project Ideas, Resources
Oct 30	MIDTERM DUE	Midterm Guide
Nov 6	Progress Check	Weekly Checklist
Nov 13	Progress Check	Weekly Checklist
Nov 20	Progress Check	Weekly Checklist
Nov 27	Final Review	Weekly Checklist
Dec 4	FINAL DUE	Final Guide



Success Formula

1. **Week 9:** Read Getting Started + Project Ideas → Choose project
2. **Week 10:** Use Midterm Guide → Submit proposal
3. **Weeks 11-14:** Follow Weekly Checklist + Getting Started → Build project
4. **Week 15:** Use Final Guide → Present & submit

Throughout: Reference AI Usage Policy, Resources, and FAQ as needed!



Pro Tips

Organization:

- Keep all documents in one folder
- Bookmark this master index
- Print Weekly Checklist (or keep it open)
- Check off tasks as you complete them

Reading Strategy:

- Don't read everything at once!
- Read what you need, when you need it
- Revisit documents as you progress
- Use FAQ for quick answers

Time Management:

- 🕒 Check Weekly Checklist every Monday
 - 📅 Mark important dates in your calendar
 - 🚦 Use "Status Check" sections to track progress
 - SOS Ask for help if you fall behind
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Ready to Start?

Your First Steps:

1. Right Now:

- ☐ Read this master index completely ✓
- ☐ Understand the document structure
- ☐ Bookmark these documents

2. Today:

- ☐ Read: Getting Started Guide (Week 9 section)
- ☐ Skim: Project Ideas & Tiers
- ☐ Check: Weekly Checklist (Week 9)

3. This Week:

- ☐ Choose your project idea
- ☐ Find teammates (if applicable)
- ☐ Start working on proposal

4. Throughout Project:

- ☐ Follow Weekly Checklist
 - ☐ Document AI usage
 - ☐ Ask questions when stuck
 - ☐ Commit to GitHub regularly
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Need Help?

Where to look first:

1. Check **FAQ** for your question
2. Review the relevant **specific guide**
3. Check **Getting Started Guide** troubleshooting
4. Ask on **Canvas discussion board**
5. Contact **instructor** (office hours or email)

Remember

These documents are your roadmap to success!

- They're organized to reduce overwhelm
- Read what you need, when you need it
- Follow the checklists
- Ask for help when stuck
- You've got this! 🚀

Document Quick Reference

1. **Midterm Guide** → Proposal requirements
2. **Final Guide** → Final submission requirements
3. **Project Ideas** → Choose your project
4. **Getting Started** → Step-by-step instructions
5. **AI Usage** → How to use AI properly
6. **Resources** → Technical setup & tools
7. **FAQ** → Quick answers
8. **Weekly Checklist** → Stay on track
9. **Quick Reference** → Print this!
10. **GitHub Templates** → Copy-paste files

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Questions? Start with the FAQ, then ask instructor!

Good luck with your project! 🎉

